

Notes

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CT Fourier Series Pairs

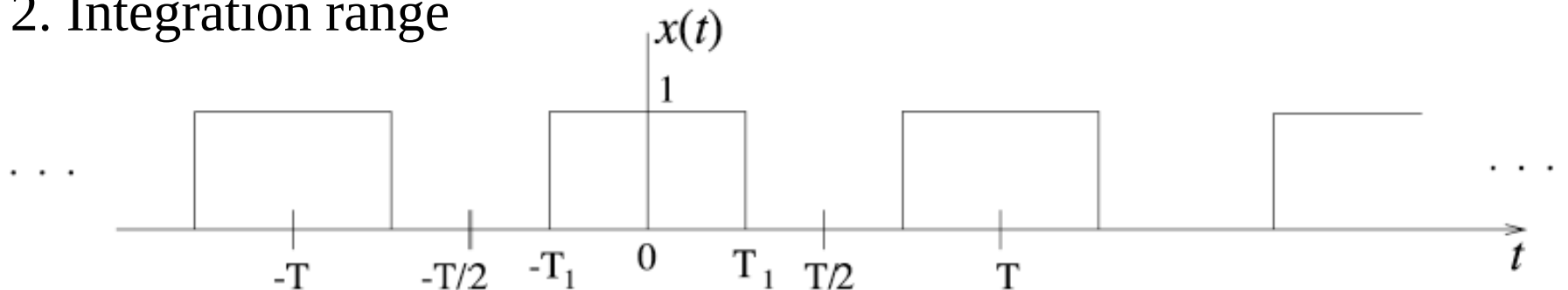
$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{j2\pi kt/T}$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

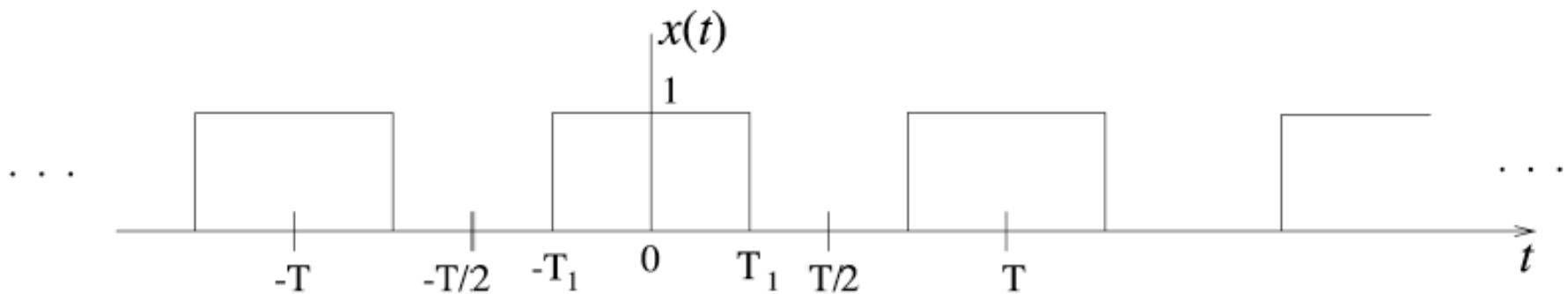
Harmonically related

1. $\omega_0 = \frac{2\pi}{T}$?

2. Integration range



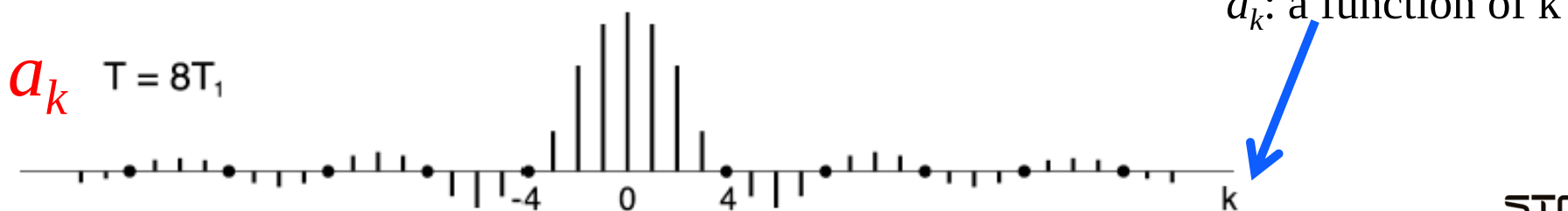
Example 3.5: Periodic Square Wave



$$a_o = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt = \frac{2T_1}{T}$$

$$k \neq 0 \quad a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt$$

$$(\omega_o = \frac{2\pi}{T}): \quad = -\frac{1}{jk\omega_o T} e^{-jk\omega_0 t} \Big|_{-T_1}^{T_1} = \frac{\sin(k\omega_o T_1)}{k\pi}$$



CT Fourier Series Property

- Linearity $x(t) \longleftrightarrow a_k, y(t) \longleftrightarrow b_k \Rightarrow \alpha x(t) + \beta y(t) \longleftrightarrow \alpha a_k + \beta b_k$
- Conjugate Symmetry

$$x(t) \text{ real} \Rightarrow a_{-k} = a_k^*$$

Proof:

$$a_{-k} = \frac{1}{T} \int_T x(t) e^{jk\omega_o t} dt = \left[\frac{1}{T} \int_T x^*(t) e^{-jk\omega_o t} dt \right]^* = a_k^*$$

$$\Downarrow a_k = \text{Re}\{a_k\} + j \text{Im}\{a_k\}$$

$$\text{Re}\{a_{-k}\} + j \text{Im}\{a_{-k}\} = |a_k| e^{j\angle a_k} \quad \text{Re}\{a_k\} - j \text{Im}\{a_k\}$$

Re{a_k} is even, Im{a_k} is odd

or |a_k| is even, ∠a_k is odd

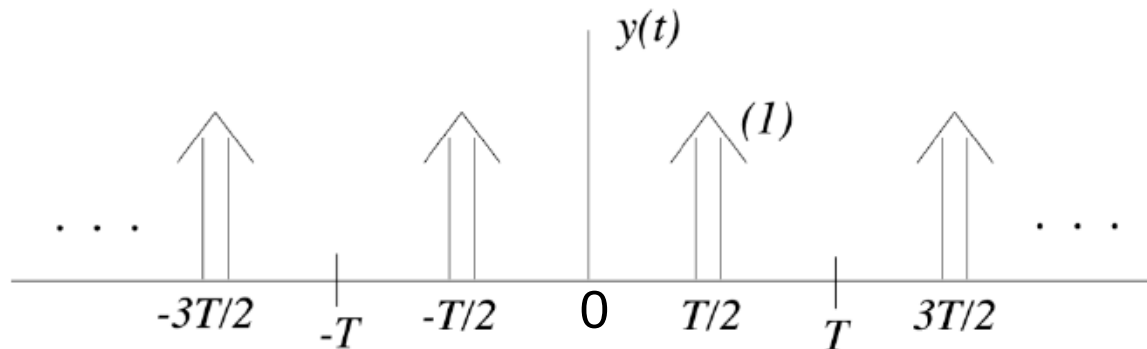
- Time shift $x(t) \longleftrightarrow a_k$
 $x(t - t_o) \longleftrightarrow a_k e^{-jk\omega_o t_o} = a_k e^{-jk2\pi t_o / T}$

Introduce a linear phase shift $\propto t_o$

Example: Shift by Half Period

what is the Fourier Series of $\sum_{n=-\infty}^{+\infty} \delta(t - nT)$?

$$y(t) = x(t - T/2) \xleftrightarrow{e^{-jk\omega_0 T/2} = e^{-jk\pi}} a_k e^{-jk\pi} = (-1)^k a_k$$



$$y(t) \longleftrightarrow (-1)^k a_k \quad (a_k = \frac{1}{T} = \text{F.C. of } \sum_{n=-\infty}^{+\infty} \delta(t - nT))$$

$$\parallel$$

$$\frac{(-1)^k}{T}$$

- Time Reversal

$$x(-t) \xleftrightarrow{FS} a_{-k}$$

the effect of sign change for $x(t)$ and a_k are **identical**

Example: $x(t): \dots a_{-2} a_{-1} a_0 a_1 a_2 \dots$

$x(-t): \dots a_2 a_1 a_0 a_{-1} a_{-2} \dots$

- Time Scaling

α : positive real number

$x(\alpha t)$: periodic with period T/α and fundamental frequency $\alpha\omega_0$

$$x(\alpha t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(\alpha\omega_0)t}$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

a_k **unchanged**, but $x(\alpha t)$ and each harmonic component are different

Cont.

- How about $x(t)$ is real, and even?

$$\begin{array}{lcl}
 x(t) = x(-t) & & \\
 x(-t) \xleftrightarrow{FS} a_{-k} & \xrightarrow{\quad} & a_k = a_{-k} \\
 x(t) \text{ is real} & \xrightarrow{\quad} & a_{-k} = a_k^* \\
 & & \text{Re}\{a_k\} \text{ is even} \\
 & & \text{Im}\{a_k\} \text{ is odd}
 \end{array}
 \xrightarrow{\quad}
 \begin{array}{l}
 \text{Im}\{a_k\} \text{ is } 0 \\
 \text{or } a_k \text{ is real, even}
 \end{array}$$

- How about $x(t)$ is real, and odd?

$$\begin{array}{lcl}
 x(t) = -x(-t) & & \\
 x(-t) \xleftrightarrow{FS} a_{-k} & \xrightarrow{\quad} & a_k = -a_{-k} \\
 & & \text{Re}\{a_k\} \text{ is } 0 \\
 & & \text{or } a_k \text{ is imaginary, odd}
 \end{array}$$

- Multiplication Property**

$x(t) \longleftrightarrow a_k, y(t) \longleftrightarrow b_k$ (Both $x(t)$ and $y(t)$ are periodic with the same period T)

$$x(t) \cdot y(t) \longleftrightarrow c_k = \sum_{l=-\infty}^{+\infty} a_l b_{k-l} = a_k * b_k$$

Proof:

$$\underbrace{\sum_l a_l e^{jl\omega_o t}}_{x(t)} \cdot \underbrace{\sum_m b_m e^{jm\omega_o t}}_{y(t)} = \sum_{l,m} a_l b_m e^{j(l+m)\omega_o t} \xrightarrow{l+m=k} \sum_k \left[\underbrace{\sum_l a_l b_{k-l}}_{c_k} \right] e^{jk\omega_o t}$$

- Parseval Relation**

$$\underbrace{\frac{1}{T} \int_T |x(t)|^2 dt}_{\text{Average signal power}} = \sum_{k=-\infty}^{+\infty} \underbrace{|a_k|^2}_{\substack{\text{Power in the} \\ k_{th} \text{ harmonic}}}$$

Power is the same whether measured in the time-domain or the frequency-domain

More

- Frequency shifting

$$e^{jM\omega_0 t} x(t) \longleftrightarrow a_{k-M}$$

Note:

$x(t)$ is periodic with fundamental frequency ω_0

$$x(t) \longleftrightarrow a_k$$

- Differentiation

$$\frac{dx}{dt} \longleftrightarrow jk\omega_0 a_k$$

- Integration

$$\int_{-\infty}^t x(t) dt \longleftrightarrow \left(\frac{1}{jk\omega_0} \right) a_k$$

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left. \begin{array}{l} x(t) \\ y(t) \end{array} \right\} \begin{array}{l} \text{Periodic with period } T \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/T \end{array}$	$\begin{array}{l} a_k \\ b_k \end{array}$
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a_k^*
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk \frac{2\pi}{T} a_k$
Integration		$\int_{-\infty}^{\infty} x(t) dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a_{-k}^* \\ \operatorname{Re}\{a_k\} = \operatorname{Re}\{a_{-k}\} \\ \operatorname{Im}\{a_k\} = -\operatorname{Im}\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \operatorname{Ev}\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \operatorname{Od}\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\begin{array}{l} \operatorname{Re}\{a_k\} \\ j\operatorname{Im}\{a_k\} \end{array}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Example 3.9

Suppose we are given the following facts about a signal $x(t)$

1. $x(t)$ is a real signal
2. $x(t)$ is periodic with period $T=4$, and it has Fourier series coefficients a_k
3. $a_k = 0$ for $|k| > 1$
4. The signal with Fourier coefficients $b_k = e^{-j\pi k/2} a_{-k}$ is odd
5. $\frac{1}{4} \int_4 |x(t)|^2 dt = 1/2$

1. $x(t)$ is a real signal
 2. $x(t)$ is periodic with period $T=4$, and it has Fourier series coefficients a_k
 3. $a_k=0$ for $|k|>1$
 4. The signal with Fourier coefficients $b_k = e^{-j\pi k/2} a_{-k}$ is odd
 5. $\frac{1}{4} \int_4 |x(t)|^2 dt = 1/2$
-

From 2), $\omega_0=2\pi/T=\pi/2$

From 3), $x(t)=a_0+a_1 e^{j\pi t/2} + a_{-1} e^{-j\pi t/2}$

From 1), $a_{-k} = a_k^*$

From 4) and 1), suppose $y(t) \longleftrightarrow b_k$, $y(t)$ is a real and odd signal

$\therefore b_k$ is imaginary, and odd

$\therefore b_0=0, b_1=-b_{-1}$, and $b_k=0$ for $|k|>1$ (why?)

$\therefore a_0=0, a_1=e^{-j\pi/2} b_{-1} = -jb_{-1} = jb_1$

$$a_{-1}=(a_1)^* = -jb_1$$

From 5), $|a_1|^2 + |a_{-1}|^2 = 1/2$

$\therefore b_1=j/2$ or $-j/2$

Periodicity Properties of DT Complex Exponentials

- $x[n]$ - periodic with fundamental period N , fundamental frequency

$$x[n + N] = x[n] \quad \text{and} \quad \omega_o = \frac{2\pi}{N}$$

$$n = \dots, -1, 0, 1, 2, 3, \dots$$

- For DT **complex exponentials**, signals are periodic only when $\omega_0 N = k \cdot 2\pi$, $k = 0, \pm 1, \pm 2, \dots$

$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)} \rightarrow e^{j\omega_0 N} = 1 \rightarrow \omega_0 N = k \cdot 2\pi$$

- For DT complex exponentials, signals with frequencies ω_0 and $\omega_0 + k \cdot 2\pi$ are identical.
$$e^{j(\omega_0 + k \cdot 2\pi)n} = e^{j\omega_0 n} \cdot e^{jk \cdot 2\pi n} = e^{j\omega_0 n}$$
 - We need only consider a frequency interval of length 2π , and in most cases, we use the interval: $0 \leq \omega_0 < 2\pi$, or $-\pi \leq \omega_0 < \pi$

Cont.

- $e^{j\omega_0 n}$ does **not** have a continually increasing rate of oscillation as ω_0 is increased in magnitude.

low-frequency (slowly varying): ω_0 near $0, 2\pi, \dots$, or $2k \cdot \pi$

high-frequency (rapid variation): ω_0 near $\pm \pi, \dots$, or $(2k+1) \cdot \pi$

$$e^{j(2k+1)\pi n} = e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

Cont.

- Harmonically-related **DT** complex exponential set

$$\{\phi_k[n] = e^{jk(\frac{2\pi}{N})n}, \quad k = 0, \pm 1, \pm 2, \dots\}$$

with common period N , i.e. $\phi_{k+N}[n] = \phi_k[n]$

$$\phi_{k+N}[n] = e^{j(k+N)(\frac{2\pi}{N})n} = e^{jk(\frac{2\pi}{N})n} \cdot e^{j2\pi n} = e^{jk(\frac{2\pi}{N})n} = \phi_k[n]$$

or there are **only N distinct signals in the set**

- So we *could* just use $e^{j0\omega_0 n}, e^{j1\omega_0 n}, e^{j2\omega_0 n}, \dots, e^{j(N-1)\omega_0 n}, k=0, \dots, N-1$
- However, it is often useful to allow the choice of N consecutive values of k to be *arbitrary* (E.g. choose $\{-(N-1)/2, (N-1)/2\}$ if N is odd and $x[n]$ has definite parity).

DT Fourier Series Representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$\sum_{k=\langle N \rangle}$ = Sum over *any* N consecutive values of k

— This is a *finite* series

$\{a_k\}$ - Fourier (series) coefficients

Questions:

- 1) What DT periodic signals have such a representation?
- 2) How do we find a_k ?

Answer to Question #1:

Any DT periodic signal has a Fourier series representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}$$

$\Downarrow$

$n=0$

$$x[0] = \sum_{k=\langle N \rangle} a_k$$

$n=1$

$$x[1] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0}$$

$n=2$

$$x[2] = \sum_{k=\langle N \rangle} a_k e^{j2k\omega_0}$$

$\vdots$

$\vdots$

$n=N-1$

$$x[N-1] = \sum_{k=\langle N \rangle} a_k e^{j(N-1)k\omega_0}$$

N equations for N unknowns, $a_0, a_1, \dots, a_{N-1}$

Answer to Question #2: A More Direct Way to Solve for a_k

Finite geometric series

$$\sum_{n=0}^{N-1} \alpha^n = \begin{cases} N & , \alpha=1 \\ \frac{1-\alpha^N}{1-\alpha} & , \alpha \neq 1 \end{cases}$$

$$\Downarrow \boxed{\alpha = e^{jk\omega_0} = e^{jk2\pi/N}}$$

$$\omega_o = \frac{2\pi}{N}$$

$$\sum_{n=\langle N \rangle} e^{jk\omega_0 n} = \sum_{n=0}^{N-1} (e^{jk\omega_0})^n = \sum_{n=0}^{N-1} (e^{jk2\pi/N})^n$$

$$= \begin{cases} N & , k=0, \pm N, \pm 2N \\ \frac{1-e^{jk\omega_0 N}}{1-e^{jk\omega_0}} = 0 & , \text{otherwise} \end{cases}$$

$$= N\delta[k - mN]$$



So, from

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}$$

$$\Downarrow \sum_{n=\langle N \rangle} e^{-jm\omega_0 n} \text{ to both sides}$$

— Taking an “inner product”
of $x[n]$ and $e^{-jm\omega_0 n}$

$$\sum_{n=\langle N \rangle} x[n] e^{-jm\omega_0 n} = \sum_{n=\langle N \rangle} \left(\sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \right) e^{-jm\omega_0 n}$$

$$= \sum_{k=\langle N \rangle} a_k \underbrace{\left(\sum_{n=\langle N \rangle} e^{j(k-m)\omega_0 n} \right)}_1$$

$N\delta[k-m]$ — orthogonality

$k=m \rightarrow N$

$k \neq m \rightarrow 0$

$$= Na_m$$

$$a_m = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jm\omega_0 n}$$

DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

Different
from CT
Fourier
series

Cont.

- We only use **N consecutive values** of a_k in the synthesis equation.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \quad (\text{Synthesis equation})$$

- a_k can be defined for **all integers k** , and we have $a_{k+N} = a_k$

$$x[n] = a_0 \phi_0[n] + a_1 \phi_1[n] + \cdots + a_{N-1} \phi_{N-1}[n]$$

$$x[n] = a_1 \phi_1[n] + \cdots + a_{N-1} \phi_{N-1}[n] + a_N \phi_N[n]$$

- ◆ Since $x[n] = x[n+N]$ and $a_{k+N} = a_k$, there are only N pieces of information, whether in the time-domain or the coefficient-domain.
- ◆ Unique for DT

Example #1: Sum of a pair of sinusoids

$$x[n] = \cos(\pi n / 8) + \cos(\pi n / 4 + \pi / 4)$$

— periodic with period $N = ?$

$$x[n] = \frac{1}{2} [e^{j\omega_0 n} + e^{-j\omega_0 n}] + \frac{1}{2} [e^{j\pi/4} e^{j2\omega_0 n} + e^{-j\pi/4} e^{-j2\omega_0 n}]$$

$\Downarrow$

$$a_0 = 0$$

$$a_{15} = a_{-1+16} = a_{-1} = 1/2$$

$$a_1 = 1/2$$

$$a_{-1} = 1/2$$

$$a_{66} = a_{2+4 \times 16} = a_2 = e^{j\pi/4}/2$$

$$a_2 = e^{j\pi/4}/2$$

$$a_{-2} = e^{-j\pi/4}/2$$

$$a_3 = 0$$

$$a_{-3} = 0$$

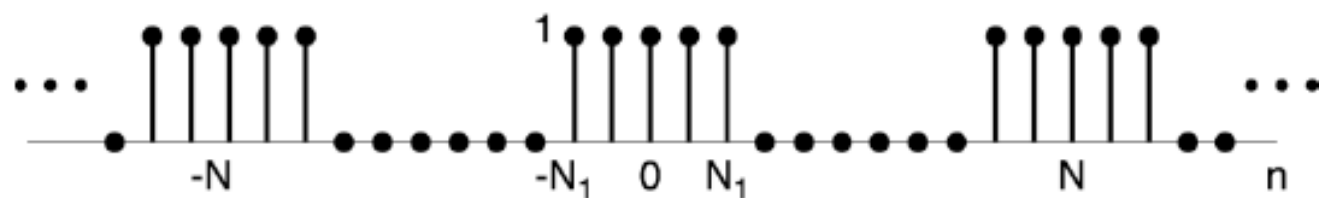
$\vdots$

$$\begin{aligned} \cos(x) &= \operatorname{Re}(e^{jx}) = \frac{1}{2} (e^{jx} + e^{-jx}) \\ \sin(x) &= \operatorname{Im}(e^{jx}) = \frac{1}{2} (e^{jx} - e^{-jx}) \end{aligned}$$

Example 3.12

DT Square wave

Period=?



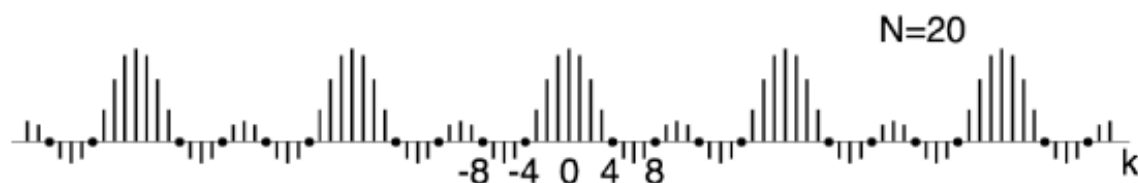
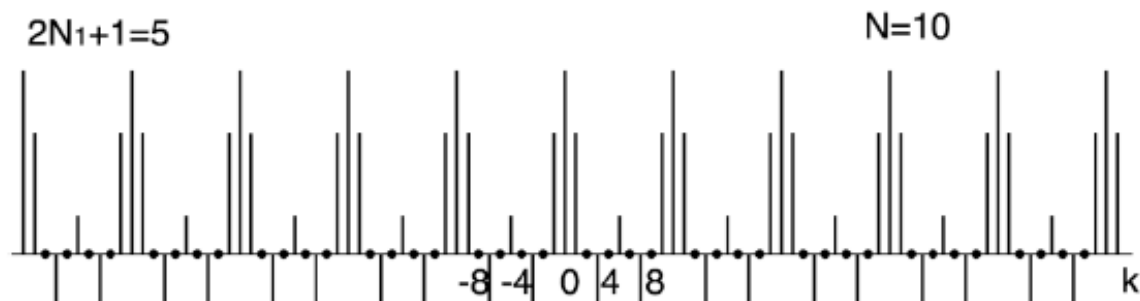
$$a_0 = \frac{1}{N} \sum_{n=-N_1}^{N_1} x[n] = \frac{(2N_1 + 1)}{N} = a_N = a_{-N} = a_{6N} = \dots$$

For $k \neq \text{multiple of } N$:

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk\omega_0 n} \stackrel{n=m-N_1}{=} \frac{1}{N} \sum_{m=0}^{2N_1} e^{-jk\omega_0 (m-N_1)} \\ &= \frac{1}{N} e^{jk\omega_0 N_1} \sum_{m=0}^{2N_1} (e^{-jk\omega_0})^m = \frac{1}{N} e^{jk\omega_0 N_1} \frac{1 - e^{-jk\omega_0 (2N_1 + 1)}}{1 - e^{jk\omega_0}} \\ &= \frac{1}{N} \frac{\sin[k(N_1 + 1/2)\omega_0]}{\sin(k\omega_0 / 2)} = \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2) / N]}{\sin(\pi k / N)} \end{aligned}$$

DT Square wave (continued)

$$a_k = \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2) / N]}{\sin(\pi k / N)}$$



DT Fourier Series - Convergence

- **Not** an issue, since all series are finite sums.
- $x[n]$ has only N parameters, represented by N coefficients

sum of N terms gives the exact value

– N odd

$$x[n]_M = \sum_{k=-M}^M a_k e^{jk\left(\frac{2\pi}{N}\right)n}$$

$$x[n]_M = x[n], \text{ if } M = \frac{(N-1)}{2}$$

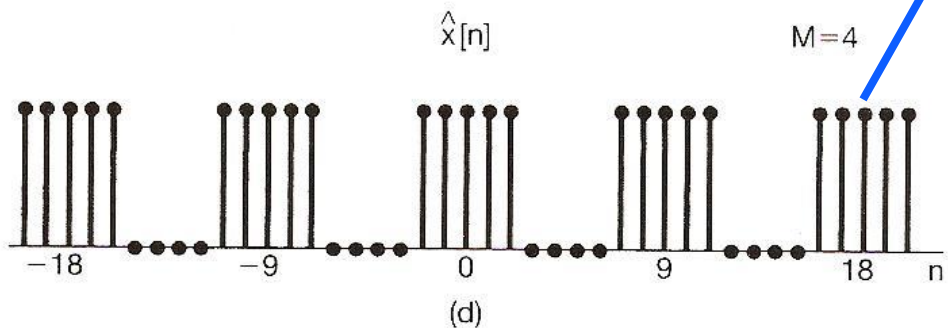
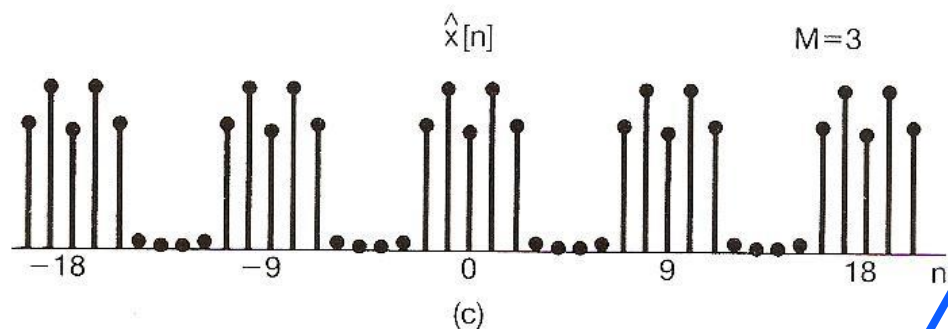
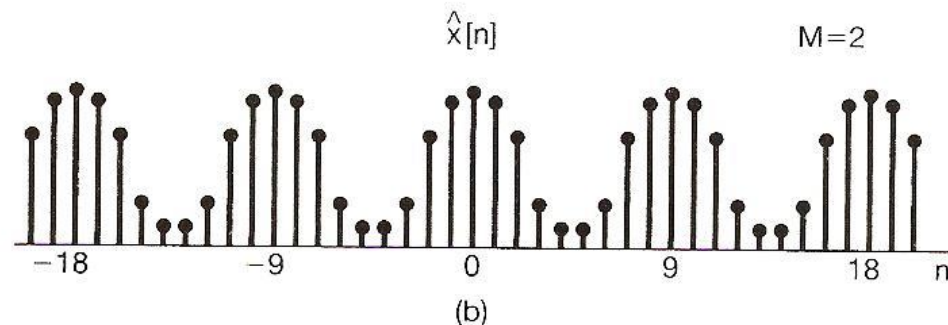
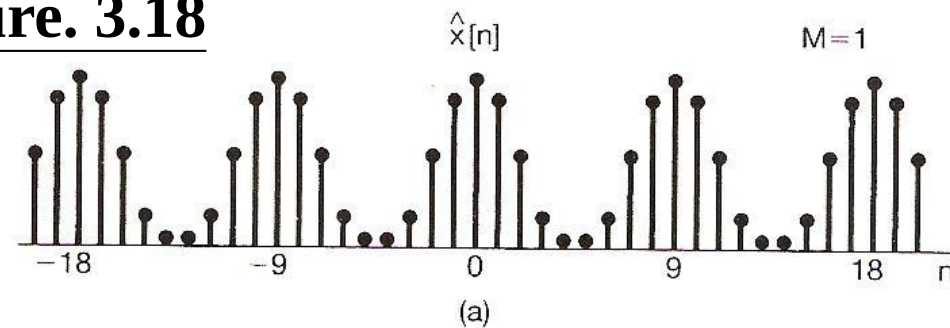
– N even

$$x[n]_M = \sum_{k=-M+1}^M a_k e^{jk\left(\frac{2\pi}{N}\right)n}$$

$$x[n]_M = x[n], \text{ if } M = \frac{N}{2}$$

See Fig. 3.18

$$N=9, 2N_1+1=5$$



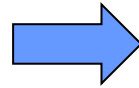
- 1) The same as original DT square wave
- 2) **No** Gibbs phenomenon, and **no** discontinuity

Figure 3.18 Partial sums of eqs. (3.106) and (3.107) for the periodic square wave of Figure 3.16 with $N = 9$ and $2N_1 + 1 = 5$: (a) $M = 1$; (b) $M = 2$; (c) $M = 3$; (d) $M = 4$.

DT Fourier Series - Property

- **Strong similarities** between the properties of DT and CT Fourier series [Comparing Table 3.2 to Table 3.1]

Example: $x[n] \longleftrightarrow a_k$



$e^{jM\omega_0 n} x[n] \longleftrightarrow b_k = ?$

$$x[n]e^{jM\omega_0 n} = \sum_{r=\langle N \rangle} a_r e^{jr\omega_0 n} e^{jM\omega_0 n}$$

$$= \sum_{k=\langle N \rangle}^{k=r+M} a_{k-M} e^{jk\omega_0 n}$$



$$b_k = a_{k-M}$$

Frequency shift
 $jM\omega_0 \rightarrow j(k-M)\omega_0$

Review **Complex Exponentials** - The Only Eigenfunctions of **Any** LTI Systems

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$\rightarrow x(t) = e^{st} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau$

$y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$

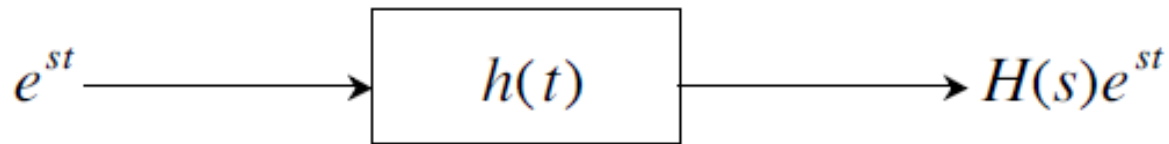
$$= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st}$$
$$= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}}$$

$\rightarrow x[n] = z^n \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{m=-\infty}^{\infty} h[m] z^{n-m}$

$$= \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m} \right] z^n$$
$$= \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}}$$

Eigenfunction Property of Complex Exponentials

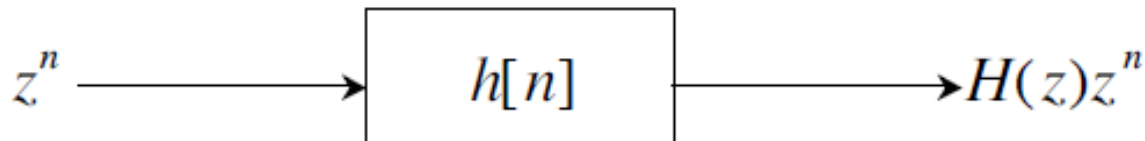
CT:



$$H(s) = \int_{-\infty}^{+\infty} h(t)e^{-st} dt$$

system function

DT:

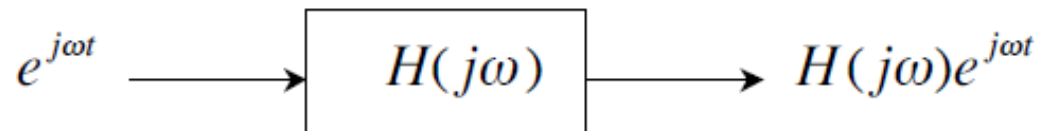


$$H(z) = \sum_{n=-\infty}^{+\infty} h[n]z^{-n}$$

system function

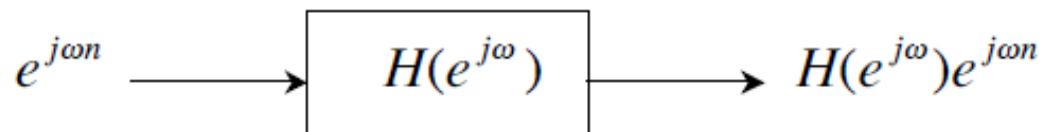
Frequency Response of an LTI System

$$(s = j\omega)$$



CT Frequency response:

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t)e^{-j\omega t} dt$$



$$(z = e^{j\omega})$$

DT Frequency response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]e^{-j\omega n}$$

Fourier Series and LTI Systems

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0) a_k}_{\text{"gain"}}$$

$$H(jk\omega_0) = |H(jk\omega_0)| e^{j\angle H(jk\omega_0)},$$

includes both amplitude & phase

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n} \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} H(e^{jk\omega_0}) a_k e^{jk\omega_0 n}$$

$$a_k \longrightarrow \underbrace{H(e^{jk\omega_0}) a_k}_{\text{"gain"}}$$

$$H(e^{jk\omega_0}) = |H(e^{jk\omega_0})| e^{j\angle H(e^{jk\omega_0})},$$

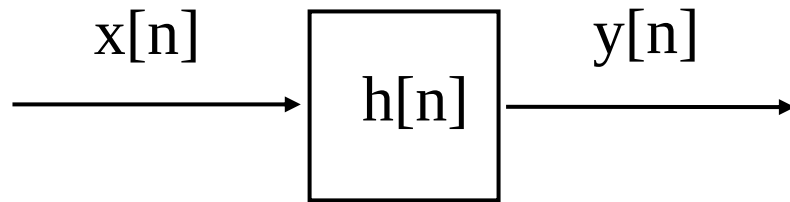
includes both amplitude & phase

The effect of the LTI system is to modify each a_k through multiplication by the value of the frequency response at the corresponding frequency.

Example 3.17

$$h[n] = \alpha^n u[n] \quad , \quad |\alpha| < 1$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) = \frac{1}{2} e^{j(\frac{2\pi}{N})n} + \frac{1}{2} e^{-j(\frac{2\pi}{N})n}$$



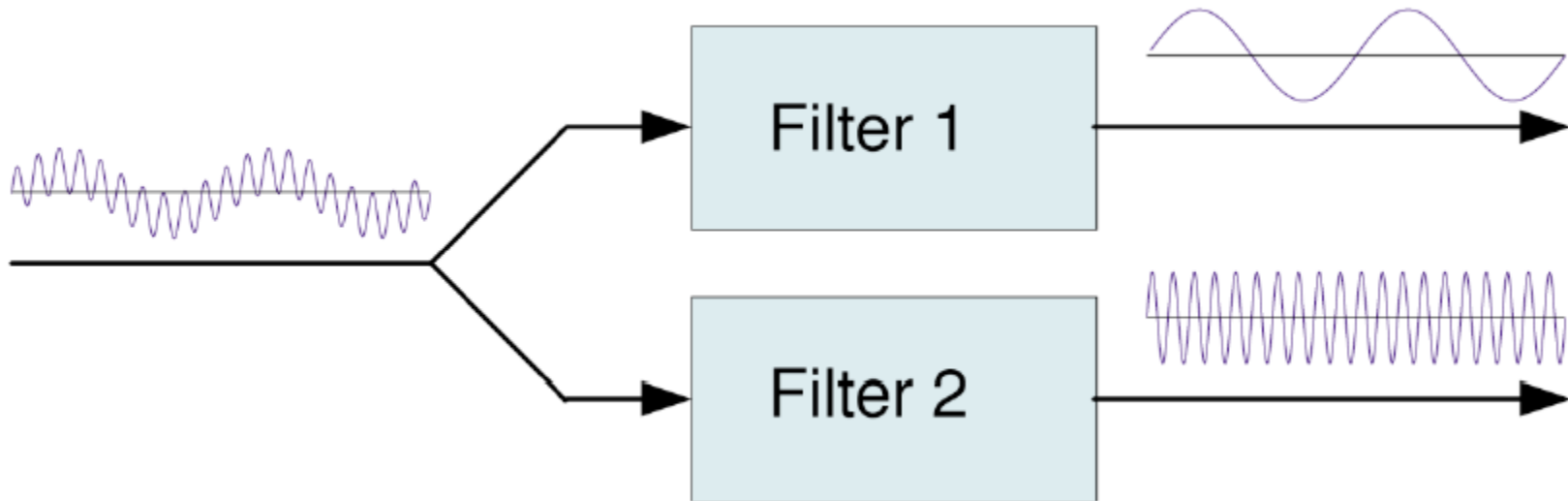
$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$\begin{aligned} y[n] &= \frac{1}{2} H\left(e^{j\frac{2\pi}{N}}\right) e^{j(\frac{2\pi}{N})n} + \frac{1}{2} H\left(e^{-j\frac{2\pi}{N}}\right) e^{-j(\frac{2\pi}{N})n} \\ &= r \cos\left(\frac{2\pi n}{N} + \theta\right) \end{aligned}$$

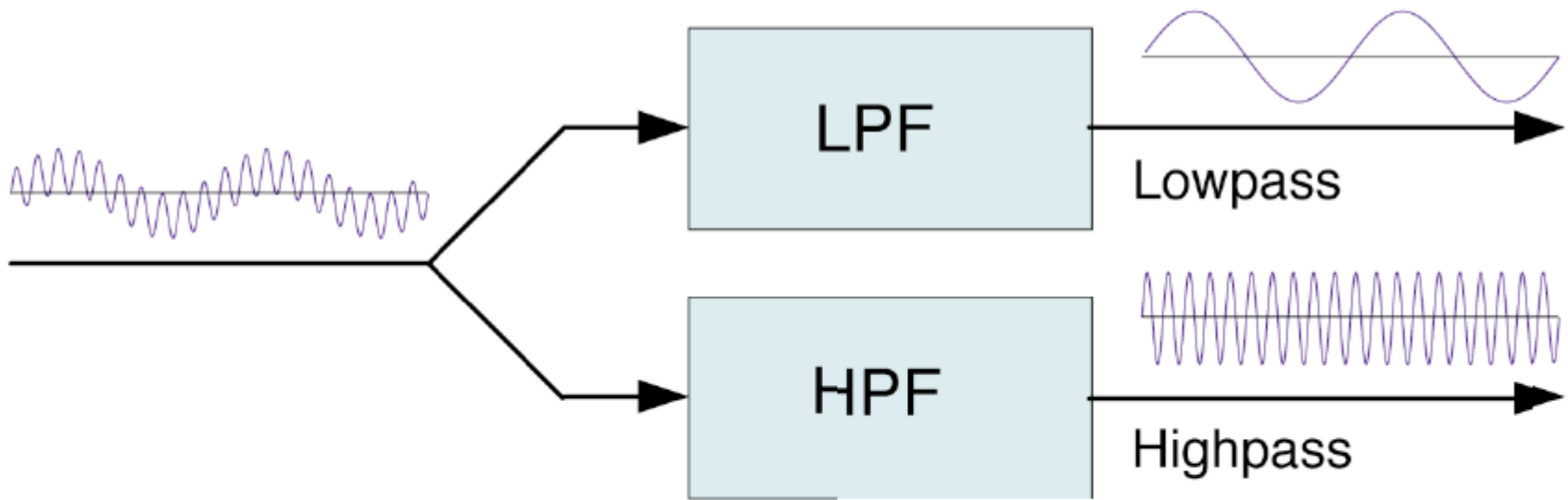
$$\text{where } re^{j\theta} = \frac{1}{1 - \alpha e^{-j\frac{2\pi}{N}}}$$

Signal Processing with Filters

- Separation of narrowband signals
 - ◆ Filter design — a main task in constructing communication systems.



Example: Using LPF and HPF to Separate Signals



Filter design becomes more challenging if the two signals are close in frequency range.

LPF: low-pass filter

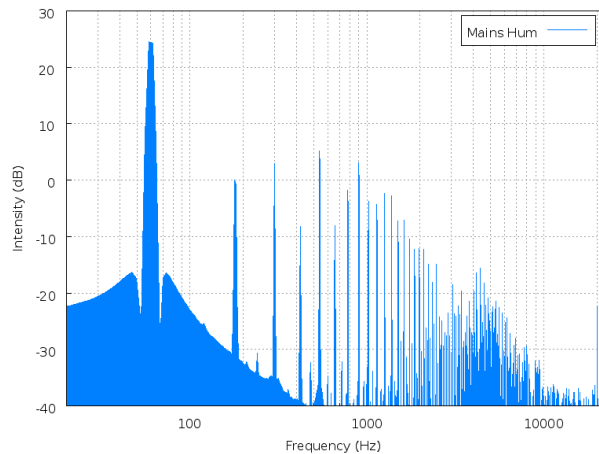
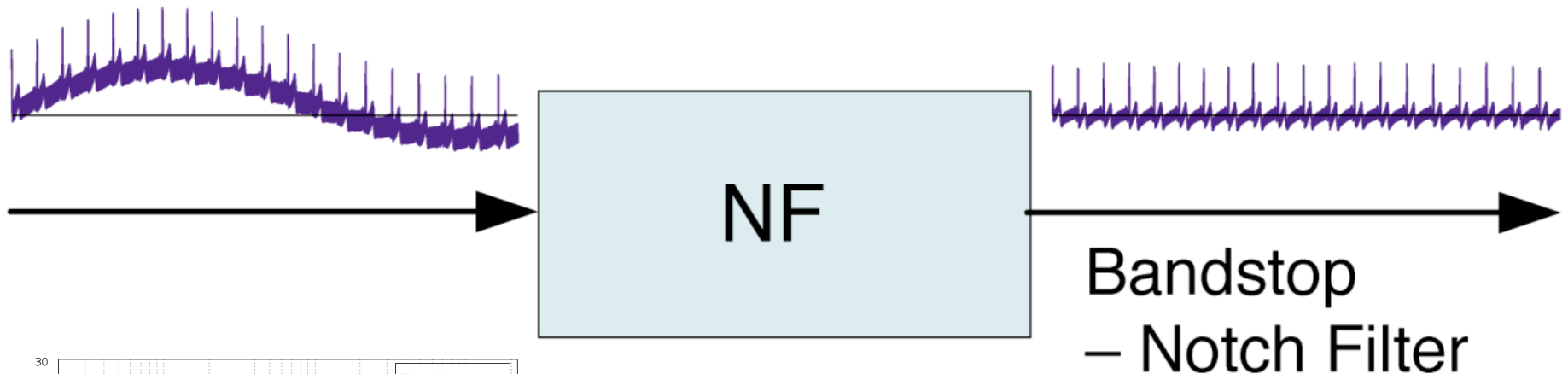
HPF: high-pass filter

Extraction of Narrowband Signals from Wideband Noise – BPF



BPF: band-pass filter

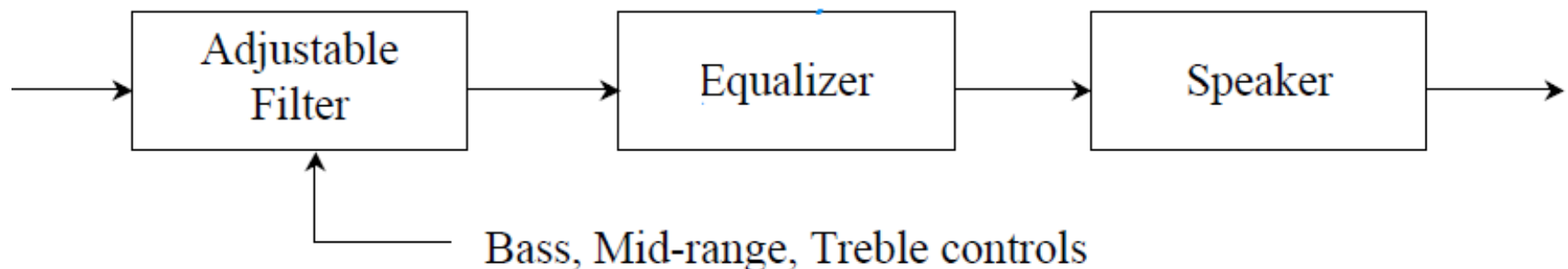
Filtering out unwanted narrowband signal, for example, 60-Hz power lines — Notch Filter



Frequency Shaping and Filtering

- By choice of $H(j\omega)$ (or $H(e^{j\omega})$) as a function of ω , we can *shape* the frequency composition of the output
 - Preferential amplification
 - Selective filtering of some frequencies

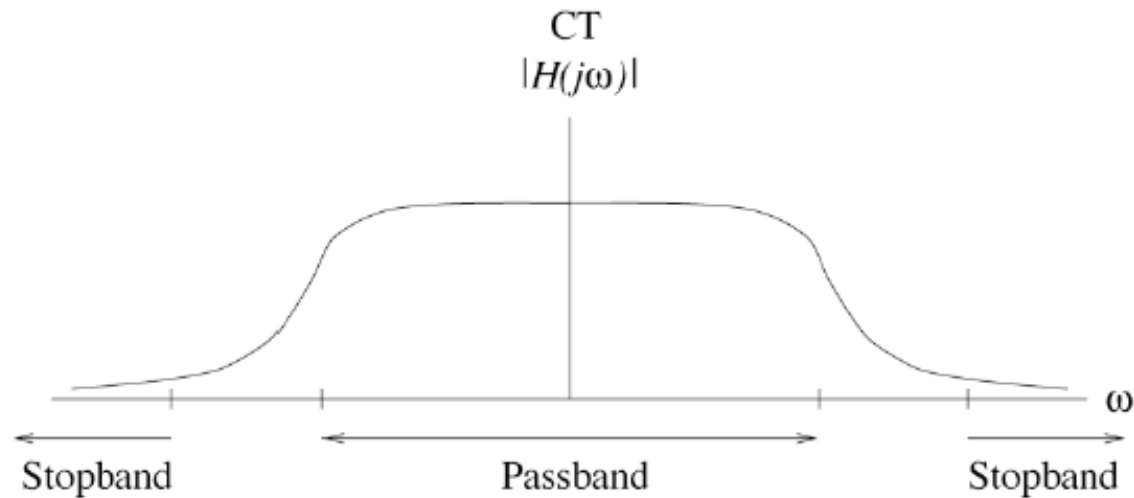
Example #1: Audio System



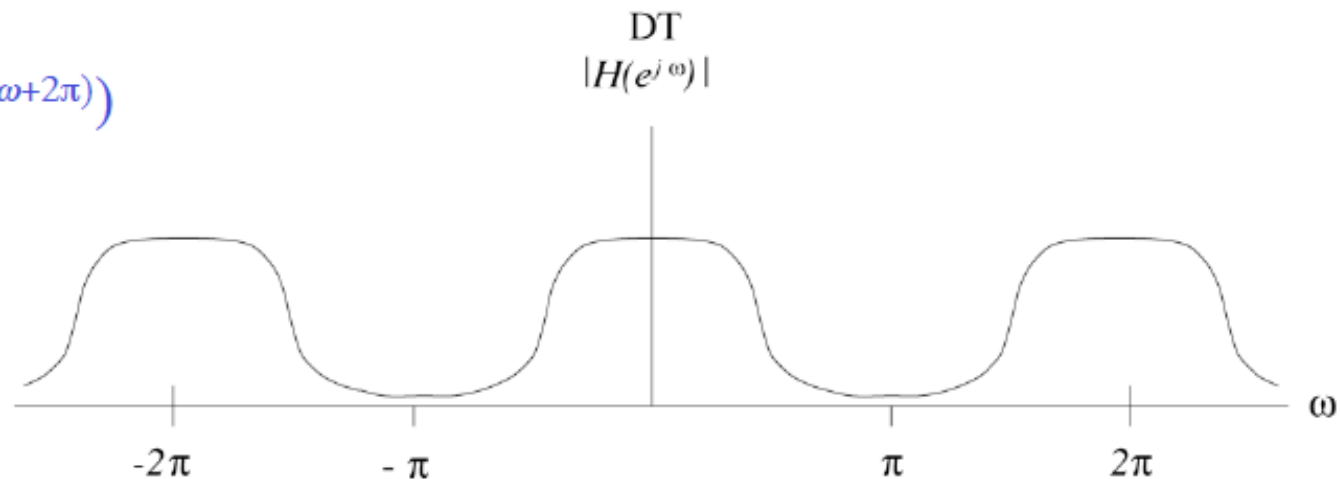
For audio signals, the amplitude is much more important than the phase.

Example #2: Frequency Selective Filters

Lowpass Filters:
Only show
amplitude here.

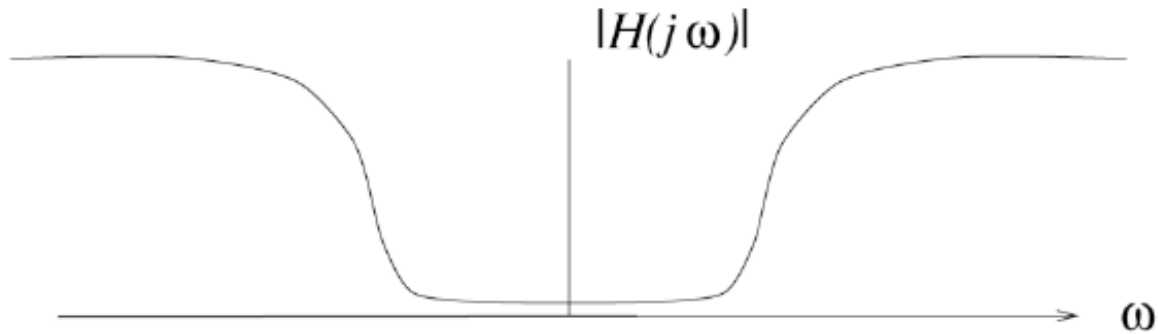


Note for DT:
 $H(e^{j\omega}) = H(e^{j(\omega+2\pi)})$

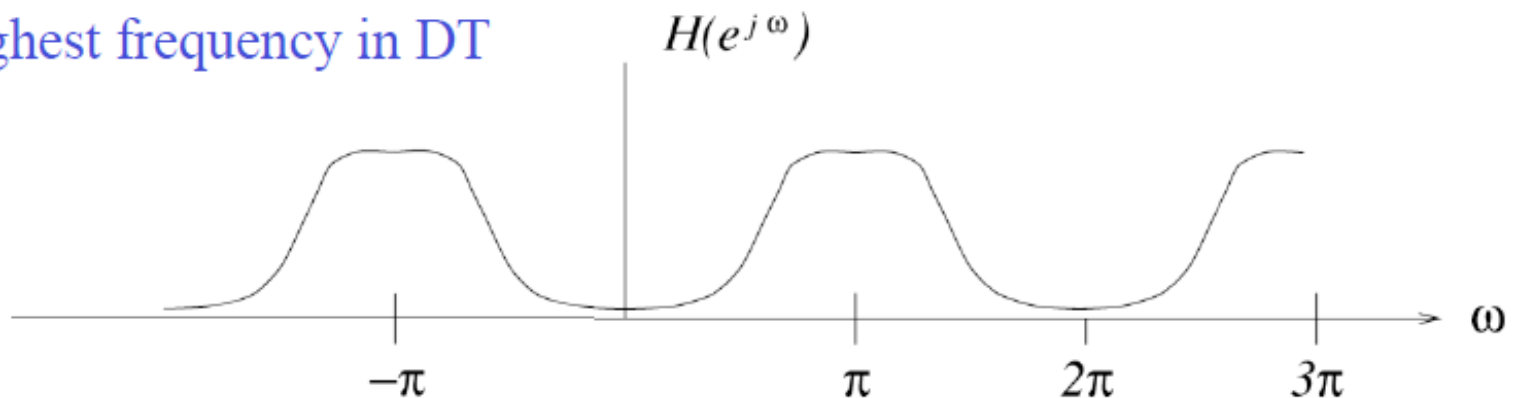


Highpass Filters

CT



DT

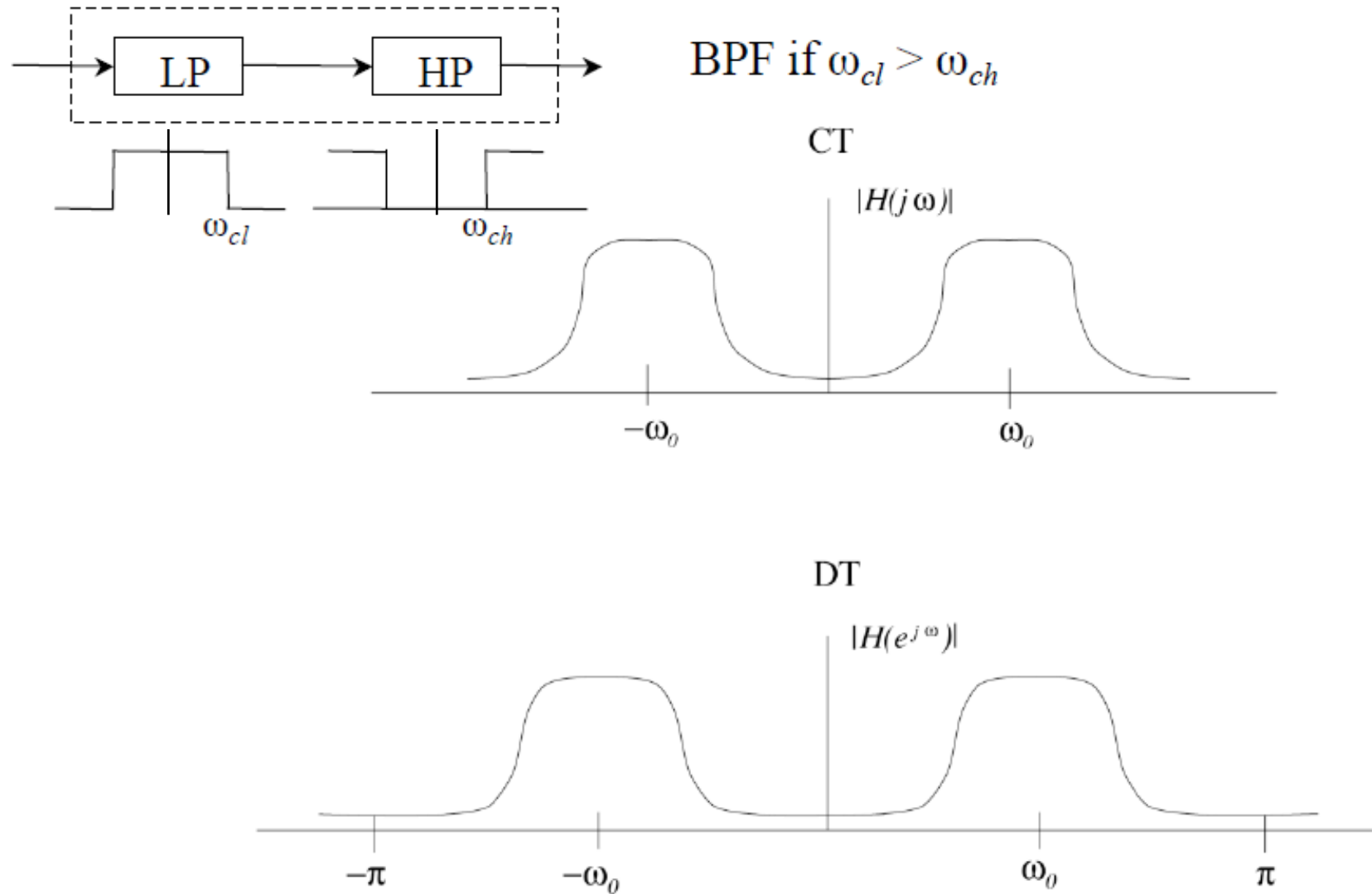


Remember:

$$(-1)^n = e^{j\pi n}$$

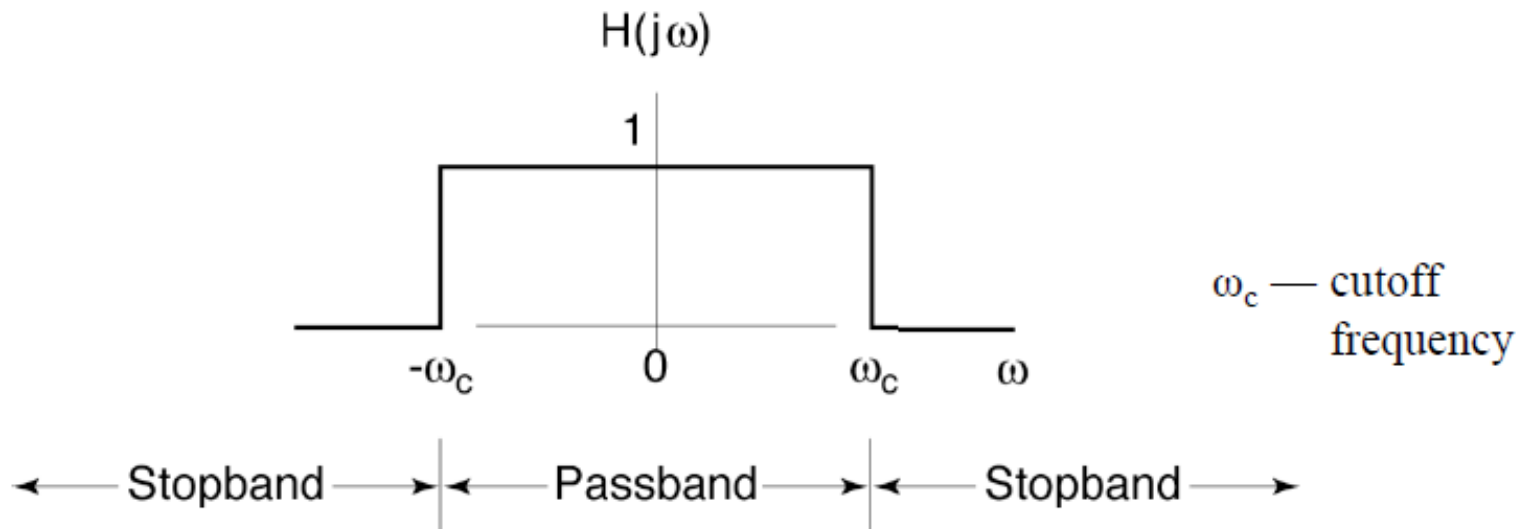
— π = highest frequency in DT

Bandpass Filters

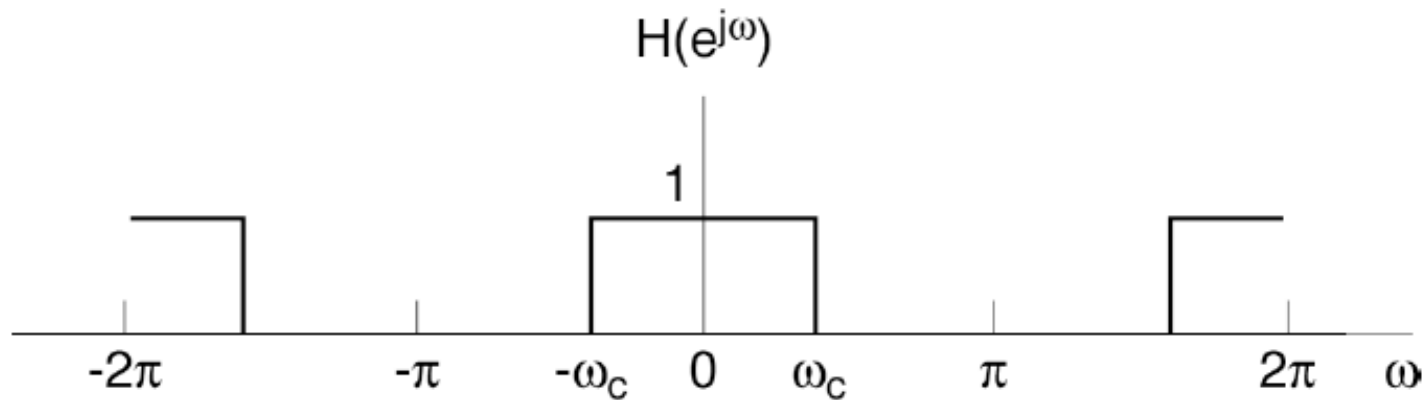


Idealized Filters

CT



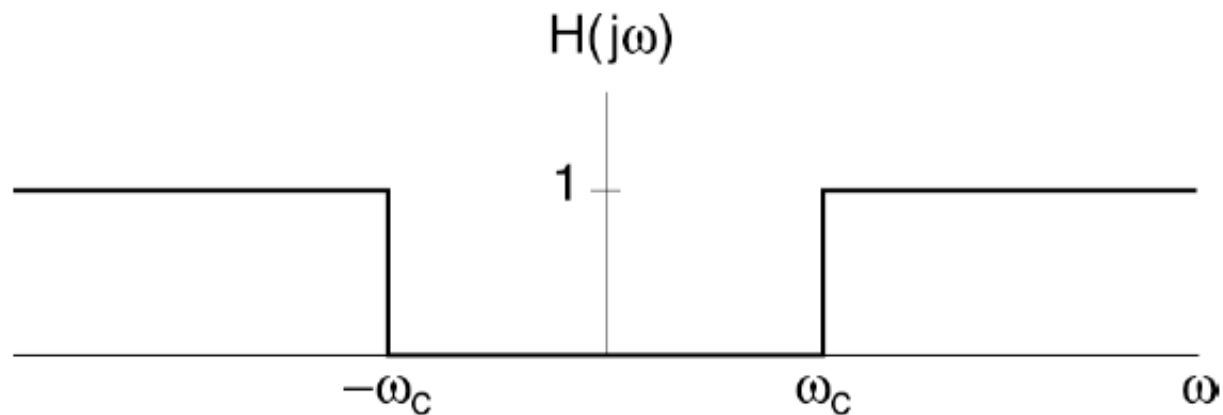
DT



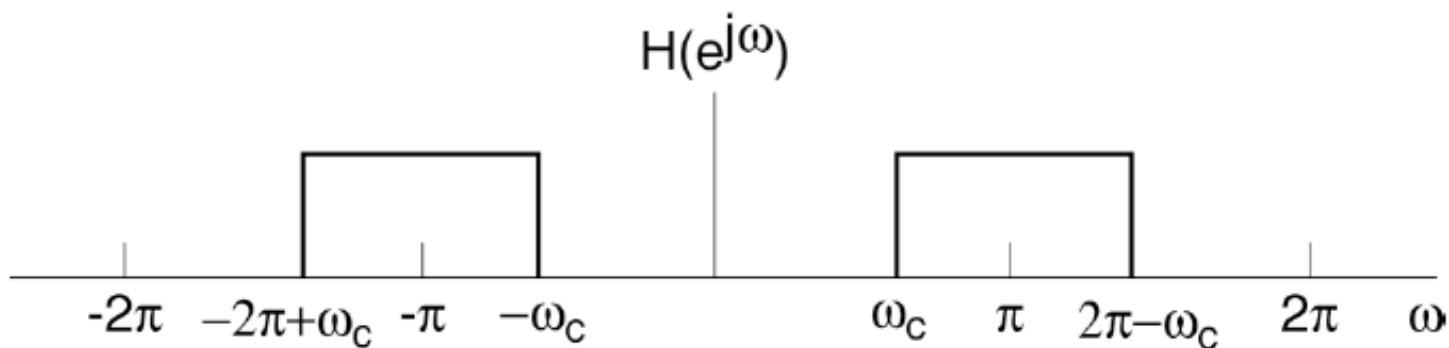
Note: $|H| = 1$ and $\angle H = 0$ for the ideal filters in the passbands,
no need for the phase plot.

Ideal Highpass Filter

CT

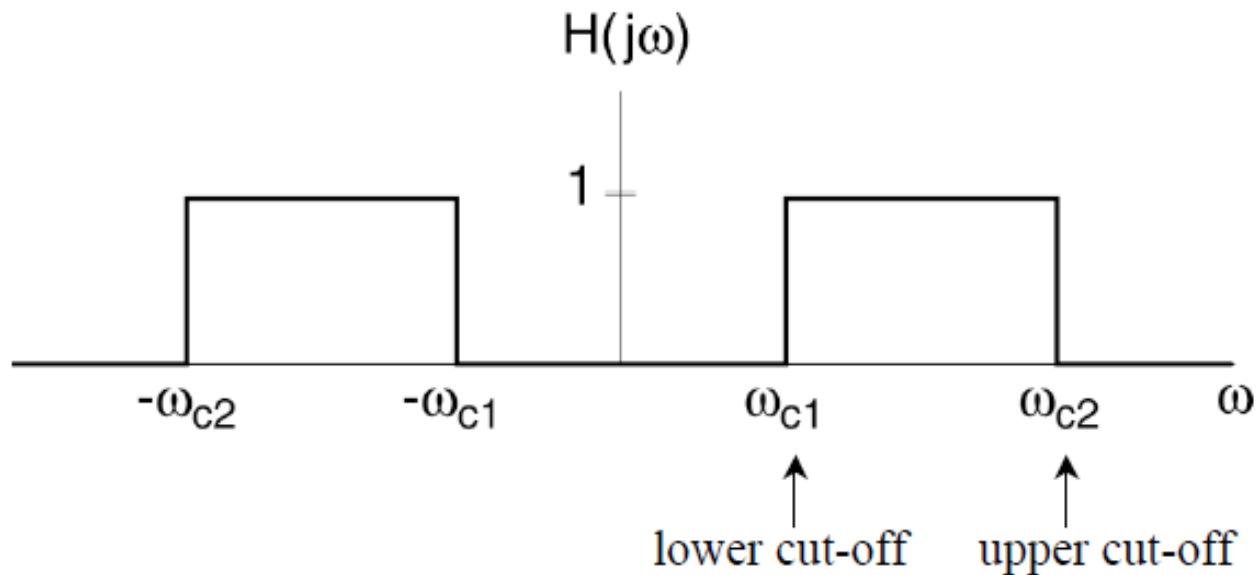


DT

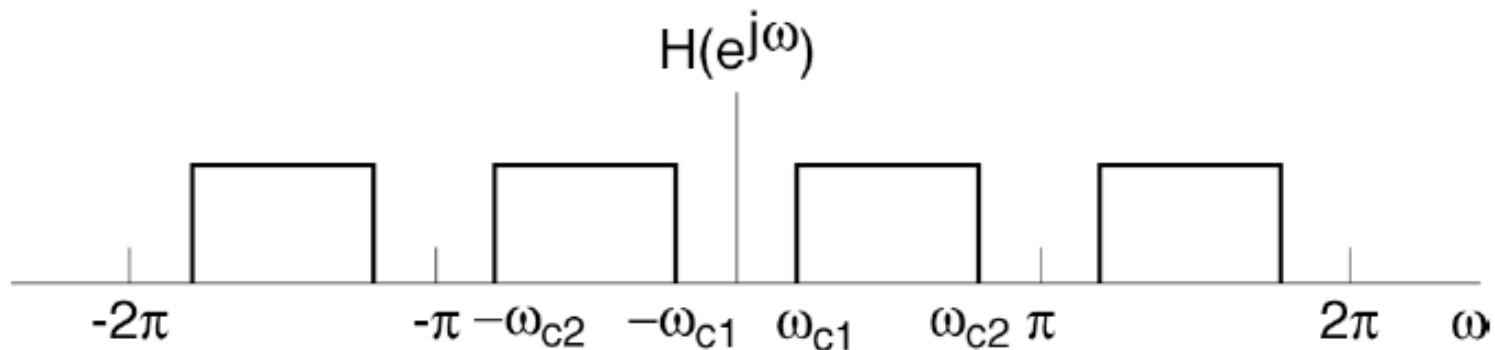


Ideal Bandpass Filter

CT



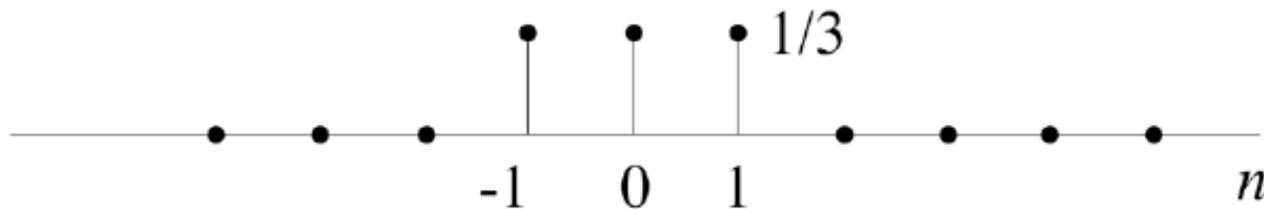
DT



Example #3: DT Averager/Smother

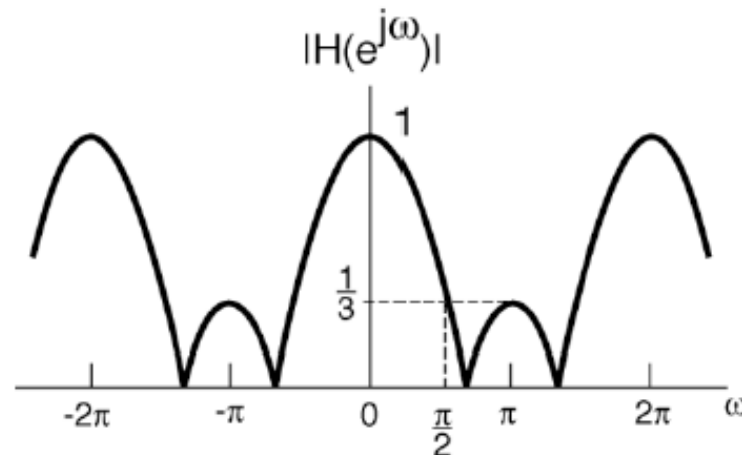
$$y[n] = \frac{1}{3}\{x[n-1] + x[n] + x[n+1]\}$$

$$h[n] = \frac{1}{3}\{\delta[n-1] + \delta[n] + \delta[n+1]\}$$



Frequency response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{3}[e^{-j\omega} + 1 + e^{j\omega}] = \frac{1}{3} + \frac{2}{3}\cos\omega$$



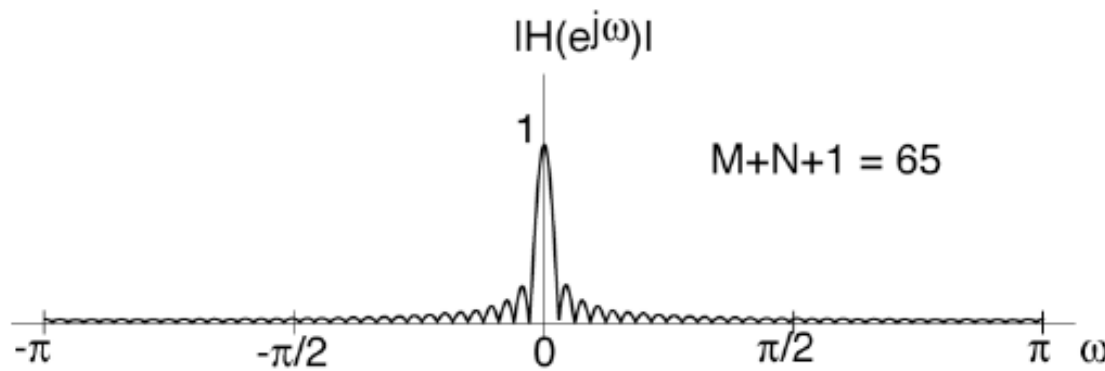
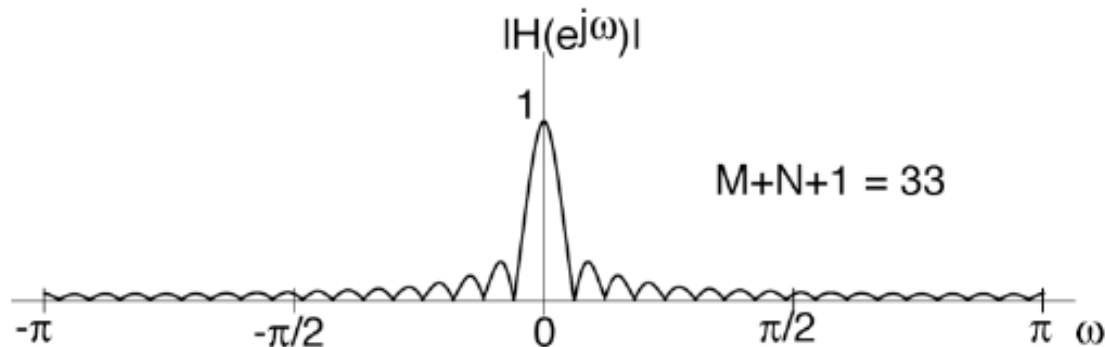
A LPF

Example #4: Nonrecursive DT (FIR) filters

$$y[n] = \frac{1}{N + M + 1} \sum_{k=-N}^M x[n - k] \longrightarrow h[n] = \frac{1}{N + M + 1} \sum_{k=-N}^M \delta[n - k]$$

Frequency response:

$$H(e^{j\omega}) = \frac{1}{N + M + 1} \sum_{k=-N}^M e^{-jk\omega} = \frac{1}{N + M + 1} e^{j\omega(N-M)/2} \frac{\sin[\omega(M + N + 1) / 2]}{\sin(\omega / 2)}$$



Rolls off at lower ω as $M+N+1$ increases

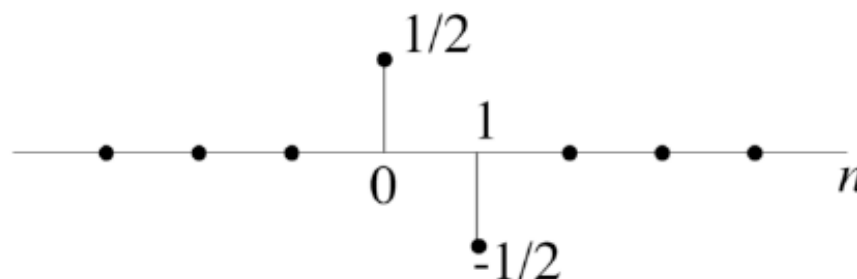
Example #5:

Simple DT “Edge” Detector

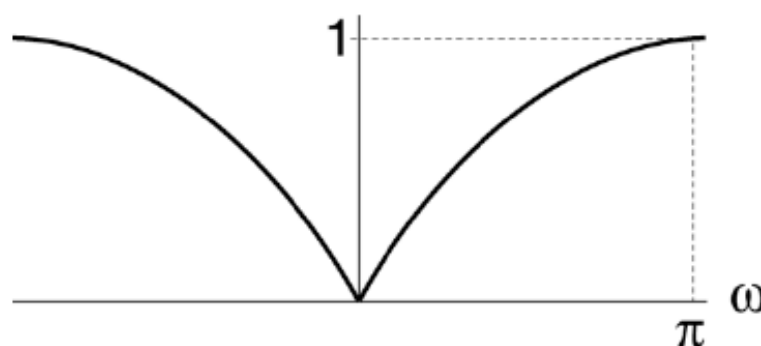
— DT 2-points “differentiator”

$$y[n] = \frac{1}{2}[x[n] - x[n-1]]$$

$$h[n] = \frac{1}{2}[\delta[n] - \delta[n-1]]$$



$|H(e^{j\omega})|$



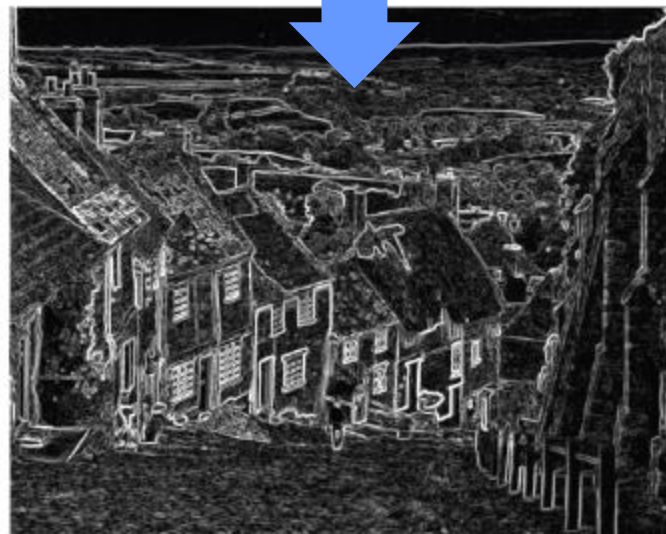
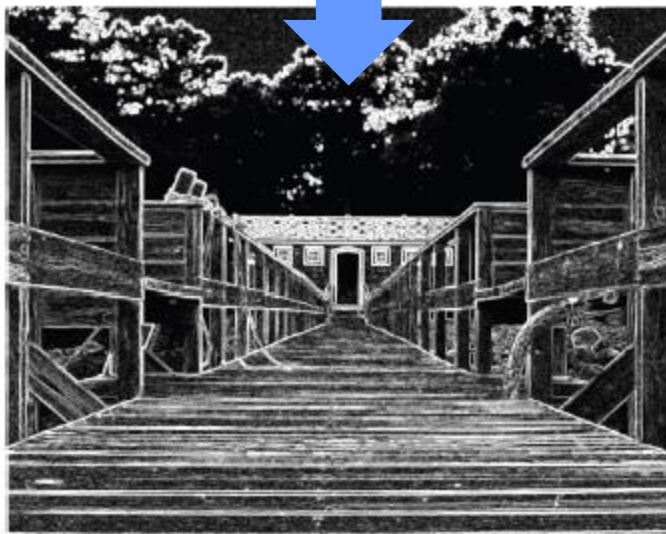
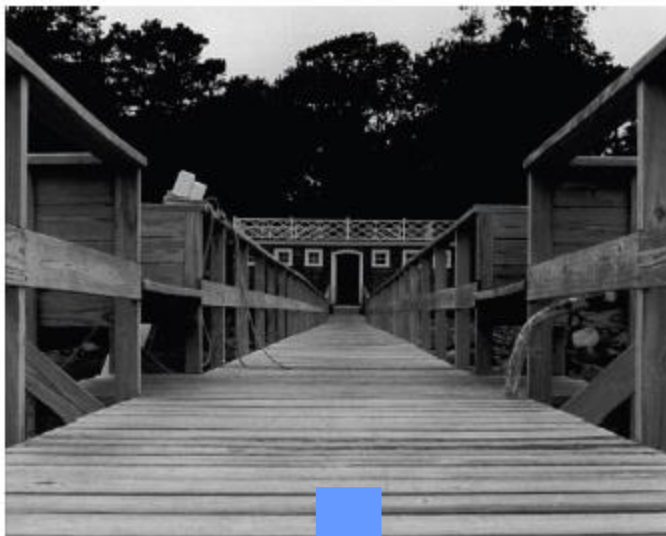
Amplifies high-frequency components

Frequency response:

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = je^{j\omega/2} \sin(\omega / 2)$$

$$|H(e^{j\omega})| = |\sin(\omega / 2)|$$

Example #6: Edge enhancement using DT differentiator



Summary

- **DT Fourier Series pair**
 - ◆ Understand the difference between CT and DT
- **Frequency response**
 - ◆ How to determine frequency response?
- **Filtering**

Problem 3.44

Suppose we are given the following facts about a signal $x(t)$

1. $x(t)$ is a real signal
2. $x(t)$ is periodic with period $T=6$, and has Fourier series coefficients a_k
3. $a_k=0$ for $k=0$ and $k>2$
4. $x(t) = -x(t-3)$
5. $\frac{1}{6} \int_{-3}^3 |x(t)|^2 dt = 1/2$
6. a_1 is a positive real number

Show that $x(t)=A \cos(Bt+C)$, and determine the values of A , B , C

1. $x(t)$ is a real signal
 2. $x(t)$ is periodic with period $T=6$, and has Fourier series coefficients a_k
 3. $a_k=0$ for $k=0$ and $k>2$
 4. $x(t) = -x(t-3)$
 5. $\frac{1}{6} \int_{-3}^3 |x(t)|^2 dt = 1/2$
 6. a_1 is a positive real number
-

From 2), $\omega_0=2\pi/T=\pi/3$

From 3),
$$x(t) = \sum_{k=-2,-1}^{1,2} a_k e^{jk\pi t/3}$$

From 4),
$$-x(t-3) = - \sum_{k=-2,-1}^{1,2} a_k e^{jk\pi(t-3)/3}$$

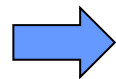
$$= - \sum_{k=-2,-1}^{1,2} a_k e^{jk\pi t/3} e^{-jk\pi} = - \sum_{k=-2,-1}^{1,2} a_k e^{jk\pi t/3} (-1)^k = \sum_{k=-2,-1}^{1,2} a_k e^{jk\pi t/3} (-1)^{k+1}$$

$$\therefore a_k = (-1)^{k+1} a_k$$

$$a_2 = -a_2$$

$$a_{-2} = -a_{-2}$$

$$\therefore a_2 = a_{-2} = 0$$



$$x(t) = a_1 e^{j\pi t/3} + a_{-1} e^{-j\pi t/3}$$

From 1), $a_{-k} = a_k^*$ $\text{Re}\{a_k\}$ is even, $\text{Im}\{a_k\}$ is odd

From 6), $a_{-1}=a_1=\text{Re}\{a_1\}$, From 5), $2|a_1|^2=1/2$, $\therefore a_1=1/2$

$$x(t) = \frac{1}{2} e^{j\pi t/3} + \frac{1}{2} e^{-j\pi t/3} = \cos(\pi t / 3)$$